

1. The asthenosphere is present just below the
(a) Stratosphere (b) Ionosphere
(c) Continental Crust (d) Upper mantle
2. What is the name of the instrument used for measurements of ozone concentrations?
(a) Long Beam UV-Photometer (b) Narrow Beam UV-Photometer
(c) Dual Beam UV-Photometer (d) Large cell UV-Photometer
3. Which oxidizing agent is known as the detergent of the atmosphere?
(a) NO (b) H₂O₂
(c) OOH radical (d) OH radical
4. Which of the following is not among the five spheres of the environment:
(a) Atmosphere (b) Biosphere
(c) Lakosphere (d) Anthrosphere
5. The Rh catalyst in the three-catalytic converter is used for
(a) production of H₂ (b) production of NO
(c) reduction of hydrocarbons (d) oxidation of CO
6. Where is the absorption maximum located in the absorption spectrum of ozone in the 200 to 350 nm wavelength range?
(a) 200-210 nm (b) 220-230 nm
(c) 250-260 nm (d) 270-280 nm
7. Which of the parameters of Lambert-Beer's law is made very large for in-situ measurement of ozone in the atmosphere?
(a) Concentration (b) Molar extinction coefficient
(c) path length (d) Molar absorptivity
8. How many nodes are there in the $\sigma_u^*(2p_z)$ of the CO molecule?
(a) None (b) One
(c) Two (d) Three
9. What is the B.O. of ¹O₂ (¹Δ)?
(a) one (b) two
(c) three (d) none of the above
10. The stretching frequency of bridging carbonyl is ——— than that of the terminal carbonyl in the metal carbonyl compounds.
(a) less (b) more
(c) equal (d) none of the above
11. Among the following metal carbonyls, which has the lowest value of stretching frequency of CO bond?
(a) Cr(CO)₆ (b) Co₂(CO)₈
(c) [Ti(CO)₆]²⁻ (d) Mn₂(CO)₁₀
12. Deoxy hemoglobin is
(a) six coordinated (b) five coordinated
(c) four coordinated (d) three coordinated

13. The binding affinity of the free Heme group for CO is about _____ times than O₂
(a) 2500 (b) 0.25 lakh
(c) 2.5 lakh (d) 25 lakh
14. Which of the following is an example of green pesticides
(a) Spinosed (b) Spinosad
(c) Spinosat (d) None of the above
15. Which of the plants is badly affected by N₂O gas?
(a) Alfalfa (b) Rice
(c) Lotus (d) None of the above
16. The typical pressure used in the Born Haber process is
(a) 200 atm (b) 2 atm
(c) 20 atm (d) 2000 atm
17. Which of the NO_x is colored at room temperature?
(a) NO (b) N₂O₅
(c) NO₂ (d) N₂O
18. How many hydroxy radicals are generated by the photocatalytic reaction of RCH₃ in photochemical smog
(a) 10,000 (b) 100
(c) 10 (d) 4
19. The role of Rh in the two-way catalytical converter is to convert
(a) NO to O₂ (b) None of the above
(c) HC to CO₂ (d) HC to H₂
20. What is the chemical name of H₂S₂O₄
(a) Thiosulphuric acid (b) Dithionic acid
(c) Dithionous acid (d) None of the above
21. Which chemical was developed as a pesticide for cotton industries in the USA?
(a) Endrin (b) DDT
(c) Toxaphene (d) None of the above
22. Which chemical is majorly responsible for the eye irritation of the photochemical smog?
(a) N₂O₅ (b) NO
(c) NO₂ (d) H₃C-(CO)-O-O-NO₂ (PAN)
23. One Dobson unit is equal to
(a) 2.7 x 10²³ ozone/sq centimeter (b) 2.7 x 10²⁰ ozone/sq meter
(c) 2.7 x 10¹⁶ ozone/sq meter (d) 2.7 x 10²⁰ ozone/sq centimeter
24. Which Cr ion coming from leather tanneries can be precipitated as a solid?
(a) Cr(III) (b) Cr(II)
(c) Cr(IV) (d) Cr(VI)
25. The chemical formula of Toxaphene is
(a) C₁₀H₁₀Cl₈ (b) C₁₂H₁₂Cl₉
(c) C₁₀H₁₀Cl₉ (d) C₁₀H₁₀Cl₁₀
26. Enzyme is a
(a) Biocatalyst (b) Non-protein material (c) Halocarbon (d) None of the options.

27. The active metal center of particulate methane monooxygenase (p-MMO) is
(a) Fe (b) Cu (c) Mo (d) Co
28. Which of the following molecules is an example of PAH?
(a) Butane (b) Naphthalene (c) Methane (d) Propane
29. Which one of the following is used as a growth substrate for *Pseudomonas methanica*?
(a) Methane (b) Ethane (c) Propane (d) TCE
30. The slope of the Lineweaver-Burk plot for enzyme-substrate reaction kinetics is
(a) v_{max} (b) $\frac{1}{v_{max}}$ (c) $\frac{K_M}{v_{max}}$ (d) $\frac{1}{S_0}$
The terms denote their conventional significances.
31. The intercept of the Lineweaver-Burk plot for enzymatic kinetics is
(a) v_{max} (b) $\frac{1}{v_{max}}$ (c) $\frac{K_M}{v_{max}}$ (d) $\frac{1}{S_0}$
32. In the presence of an inhibitor in the enzyme-substrate reaction, which of the following is correct for non-competitive enzyme inhibition?
(a) Inhibitor binds to an enzyme and reduces its capability to bind with substrate
(b) Inhibitor binds to enzyme and enzyme-substrate complex both
(c) Inhibitor binds only with the enzyme
(d) Inhibitor binds only with the enzyme-substrate complex
33. Kreb's cycle is
(a) 4-step process (b) 8-step process (c) 6-step process (d) 10-step process
34. In Kreb's cycle, which of the following enzymes transform citrate into isocitrate?
(a) α -ketoglutarate (b) Aconitase (c) Isocitrate (d) Fumarase
35. Environmental factor (E-factor) is the measurement of
(a) Total Waste (kg) / Product (kg) (b) Product (kg) / Total Waste (kg)
(c) Total Waste (kg) / Total reactant (kg) (d) Reactant (kg) / Product (kg)
36. The Atom Economy concept was developed by
(a) Prof. B.M Trost (b) Prof. Hoechst (c) Prof. Paul T. Anastas (d) Prof. John C. Warner
37. In 2002, Nature Works, a Cargill company, won a Green Chemistry award for producing
(a) Biodegradable plastic from corn (b) Atom economy
(c) Industrial ecology (d) Ibuprofen production
38. FAD is
(a) Flavin adenine dinucleotide (b) Flavin adenosine dinucleotide
(c) Flavin adenosine trinucleotide (d) Flavin adenosine dinucleotide hydrate

39. In β -oxidation of fatty acid molecules, which one of the following is produced as the final product?
(a) Fatty acyl CoA (b) β -Ketoacyl CoA (c) Acetyl-CoA (d) β -Hydroxyacyl CoA
40. Which one among the following compounds decomposes at the slowest rate?
(a) Lignin (b) Sugar (c) Protein (d) Starch
41. PCBs are
(a) Pyrroloquinoline benzenes (b) Polychlorinated biphenyls
(c) Polychlorinated butanes (d) Polychlorinated butyl ethers.
42. In Candida Maltose the conversion of alkane to 1-Fatty Alcohol is carried out by the following enzyme
(a) Fatty alcohol oxidase (b) Fatty aldehyde reductase (c) P450 enzyme (d) FAD
43. Proteins are the polymer of
(a) Sugars (b) Amino acid (c) Carboxylic acids (d) Aliphatic amines
44. The synthesis of catechol from glucose using biocatalytic has been demonstrated using the following bacteria
(a) E.Coli (b) Pseudomonas Putida (c) Candida maltose (d) Candida tropicalis
45. Ionic liquids are considered the ideal green solvents because they are
(a) Non-volatile, non-flammable and non-explosive (b) Unstable at high temperatures
(c) Volatile at room temperature (d) Explosive
46. Which is not the characteristic of a consumable product?
(a) Degradable (b) Non-bio-accumulative (c) Non-toxic (d) Recyclable
47. Biodegradation of long-chain alkane present in petroleum follows the following biodegradation pathways.
(a) Alkane $\rightarrow$ Alcohol $\rightarrow$ Aldehyde $\rightarrow$ Acid $\rightarrow$ Acetyl CoA
(b) Alkane $\rightarrow$ Alcohol $\rightarrow$ Aldehyde $\rightarrow$ Acetyl CoA $\rightarrow$ Acid
(c) Alkane $\rightarrow$ Acetyl CoA $\rightarrow$ Aldehyde $\rightarrow$ Alcohol $\rightarrow$ Acid
(d) Alkane $\rightarrow$ Alcohol $\rightarrow$ Aldehyde $\rightarrow$ Acetyl CoA $\rightarrow$ Acid
48. Industrial ecology implies
(a) Industrial waste or by-products become significant input for another process.
(b) Recycling each other's waste materials
(c) Efficient utilization of energy
(d) All the options are correct
49. Which of the following is one of the most notable examples of Industrial Ecology in practice today?
(a) The Hoover Dam (b) The Redwood Forest in California
(c) Denmark's Kalundborg Eco-industrial Par (d) The solar power plants in the Mojave Desert
50. TCEs
(a) Trichloroethylene (b) Trichloroethane (c) Trichloroethanol (d) Trichloroepoxide